Choosing a Method: A Study Map

DSC 210 Foundations of Data Science · Modules 9 to 11 (unsupervised learning, classification, regression)

supervised · unsupervised · prepare · judge and compare

1 Does your question ask you to predict one column, and do you know that column's values for at least some rows?

YES: there is a label

NO: no label

SUPERVISED LEARNING

Learn from rows where the answer (the label) is known; predict it for new rows.

2 Is the label a category or a number?

Category: CLASSIFICATION

Each prediction is right or wrong.

k-nearest-neighbors M10
The k closest training rows vote.
Watch: needs scaling; k = 1 memorizes

Decision tree M10
Yes/no questions you can read as a rule.
Watch: overfits without a max_depth

Random forest M10
Many varied trees vote; steadier than one tree.
Watch: no single readable rule

Logistic regression M11
Gives a probability; you choose the threshold.
Watch: straight-line boundary

Course example: *Which species is this penguin? Is this email spam?*

Number: REGRESSION

Each prediction is off by some amount.

Linear regression M11
A slope in real units; easy to explain.
Watch: curves; extrapolation past the data

k-NN regression M11
Average of the k closest rows; bends with the data.
Watch: needs scaling; no slope to report

Regression tree M11
A step function you can read.
Watch: overfits badly when unlimited

Random forest M11
Averages many trees; often best with many features.
Watch: importances mislead when features are correlated

Course example: *How many rings does this abalone have?*

UNSUPERVISED LEARNING

No answer key; look for structure that is already there.

2 Looking for groups, or for fewer columns?

Groups: CLUSTERING

Rows that resemble each other.

k-means M9
You choose k (elbow plot); round, similar-size groups.
Watch: fails on curved shapes

Hierarchical M9
Dendrogram shows every k; cut it afterwards.
Watch: linkage choice changes the tree

DBSCAN M9
Dense groups of any shape; labels outliers as noise.
Watch: eps is hard to choose

Course example: *Do river samples sort into seasons?*

Fewer columns: PCA

Summarize many numeric columns.

Principal components M9
A few components keep most of the variance; read the loadings to name them.
Watch: dropping components shrinks distances

Course example: *Six spending columns in one picture.*

3 Prepare before you fit

- **Clean first:** handle missing values and look for impossible ones (Module 8; the 226 mm abalone).
- **Split** into training and test sets (use `stratify` for a category label). The test set influences no decision.
- **Scale** for anything built on distance: k-NN, k-means, hierarchical, DBSCAN, PCA (and logistic regression, to compare coefficients). Trees and forests need no scaling.
- Fit the scaler on the **training** data only.
- Every hyperparameter is a choice to justify: k, max_depth, n_estimators, eps, number of clusters or components.

4 Judge the result, then compare honestly

Classification

- Baseline: always guess the most common class.
- Confusion matrix; precision, recall, F1.
- Accuracy alone misleads when one class is rare.

Regression

- Baseline: always guess the mean.
- MAE, RMSE, R^2 (negative R^2 = worse than the mean).
- Residual plot: no pattern, curve, or fan?

Unsupervised

- No accuracy to compute.
- Elbow, dendrogram gaps, agreement between methods.
- Do the cluster profiles make sense to an expert?

Choosing among models (supervised):

k-fold cross-validation on the training data → pick a model: mean score, spread, simplicity → one final score on the test set

Shuffle the folds unless the rows are in time order and you are forecasting the future.

Warning signs. Training score far above test score (overfitting: small k, deep trees) · predictions outside the range of the data (extrapolation) · clusters that appear even in random noise · a coefficient or importance read alone when features are correlated.

No Free Lunch. Averaged over all possible problems, no single method is best. Each one wins only where its assumptions fit the data, so try more than one and let validation decide.